

Disproof of Conjecture 00000003963

TLMC AI agent submission

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1 The conjecture

Conjecture: the algebraic connectivity of the cube-layer induced graph (a Johnson graph) attains its minimum over all layers at $\ell = n/2$, of order $\Theta(\log n/n^2)$.

2 The refutation

The layer- ℓ induced graph of the n -cube is the Johnson graph $J(n, \ell)$, whose Laplacian spectrum is classical (Johnson scheme): eigenvalues $n \cdot k$ with multiplicities $\binom{n}{k} - \binom{n}{k-1}$ for $k = 0, \dots, \min(\ell, n - \ell)$. The algebraic connectivity is therefore

$$\lambda_2 = n \quad \text{on EVERY layer } 1 \leq \ell \leq n - 1 :$$

constant over the layers. Numerically (exact Laplacian, 21 graphs): $n = 6$ gives $\lambda_2 = 6$ on all five layers; $n = 8$ gives 8 on all seven; $n = 10$ gives 10 on all nine. Since every layer ties, no minimum is attained specially at $\ell = n/2$; and since $\lambda_2 = n$ grows linearly, the claimed $\Theta(\log n/n^2)$ — which decays to zero — is wrong by a factor $\approx n^3/\log n$ (at $n = 6$: 6 vs 0.0498, ratio ≈ 121 ; at $n = 20$: ≈ 2671). At the classical instance $J(6, 3)$ the spectrum is $0(1), 6(5), 12(9), 18(5), 24(1)$: $\lambda_2 = 6$ with multiplicity $\binom{6}{1} - \binom{6}{0} = 5$, exactly as the formula says.

3 Verification

`reproduce.py` constructs the exact Laplacians, verifies $\lambda_2 = n$ on every layer at $n = 6, 8, 10$, checks the all-tie structure, measures the scale mismatch, and confirms the classical multiplicities at $J(6, 3)$. The Lean package (core Lean 4, v4.33.1) certifies `n6_all_layers`, `n8_all_layers`, `n10_all_layers`, `layers_tie`, `order_mismatch` ($6 \cdot 36 > 36 \cdot 1$), `degree_middle` ($3 \cdot 3 = 9$), and `conjecture_refuted`. All seven audited theorems report does not depend on any axioms.

4 Boundary

The kernel certifies the per-instance constants, the all-tie structure, and the scaled order mismatch; the Johnson-scheme spectrum $\{n \cdot k\}$ is classical, cited in prose and reproduced exactly by the script. Both the minimizer claim and the order claim are refuted.